

```
net start MongoDB
```

```
# Press Enter
```

```
mongosh
```

```
# Press Enter
```

```
use collegeDB
```

```
# Press Enter
```

```
db.students.insertMany([
```

```
{
```

```
  name: "Rahul",
```

```
  department: "CSE",
```

```
  marks: 85,
```

```
  city: "Delhi"
```

```
},
```

```
{
```

```
  name: "Anita",
```

```
  department: "ECE",
```

```
  marks: 78,
```

```
  city: "Mumbai"
```

```
},
```

```
{
```

```
  name: "Vikram",
```

```
  department: "CSE",
```

```
  marks: 92,
```

```
  city: "Delhi"
```

```
},  
  
{  
  name: "Sneha",  
  department: "ME",  
  marks: 67,  
  city: "Pune"  
},  
  
{  
  name: "Karan",  
  department: "CSE",  
  marks: 74,  
  city: "Mumbai"  
}  
])  
  
# Press Enter
```

```
db.students.find()  
  
# Press Enter
```

```
db.students.aggregate([  
  {  
    $group: {  
      _id: "$department",  
      totalStudents: { $sum: 1 }  
    }  
  }  
])  
  
# Press Enter
```

```
db.students.aggregate([
{
  $group: {
    _id: "$department",
    averageMarks: { $avg: "$marks" }
  }
}
])
# Press Enter
```

```
db.students.aggregate([
{
  $group: {
    _id: null,
    highestMarks: { $max: "$marks" }
  }
}
])
# Press Enter
```

```
db.students.aggregate([
{
  $group: {
    _id: null,
    lowestMarks: { $min: "$marks" }
  }
}
])
```

Press Enter

```
db.students.aggregate([  
  {  
    $sort: {  
      marks: -1  
    }  
  }  
])
```

Press Enter

```
db.students.aggregate([  
  {  
    $match: {  
      marks: { $gt: 75 }  
    }  
  }  
])
```

Press Enter

```
db.students.aggregate([  
  {  
    $project: {  
      _id: 0,  
      name: 1,  
      marks: 1  
    }  
  }  
])
```

```
}  
])
```

Press Enter

```
db.students.createIndex({ name: 1 })
```

Press Enter

```
db.students.createIndex({ marks: -1 })
```

Press Enter

```
db.students.createIndex({ department: 1, marks: -1 })
```

Press Enter

```
db.students.getIndexes()
```

Press Enter

```
db.students.dropIndex({ name: 1 })
```

Press Enter

```
db.students.find({ name: "Rahul" })
```

Press Enter

```
db.students.find({ name: "Rahul" }).explain("executionStats")
```

```
# Press Enter
```

```
db.students.drop()
```

```
# Press Enter
```

```
exit
```

```
# Press Enter
```